Situation Entity Annotation for German Parliamentary Debates

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Abstract

This document describes the annotation of Situation Entities in German parliamentary debates. Our guidelines are based on previous work by [Friedrich et al., 2015, Mavridou, 2016][1] and should be considered as an extension to the original guidelines. Here we focus on phenomena specific to German or to parliamentary debates.

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[1] https://www.coli.uni-saarland.de/projects/sitent/page.php?id=resources

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1 Introduction

1.1 Overview

Situation Entities have been introduced by [Smith, 2003] in her book "Modes of discourse" where the author identifies five distinct discourse modes: Narrative, Description, Report, Information, and Argument. Discourse modes can be seen as *linguistic* properties of text passages, thus contrasting a more pragmatic view that typically focusses on the genre of a text. According to Smith, discourse modes can be characterised by two features, (i) the situations they introduce (such as states, events, ...) and (ii) their manner of progression, either temporal or metaphorical. Smith's Situation Entity (SE) types provide a classification for semantic clause types, covering eventualities (States, Events, Reports), general statives (Generics, Generalising sentences) and abstract entities (Facts, Propositions). Each discourse mode can be identified by its distinct pattern of Situation Entity types that are predominant in this particular mode.

Situation Entities (SE) operate on the clause level and are evoked by the main verb and its arguments. They can refer to situations in the world as well as to more abstract descriptions of kinds or abstract individuals (Generics). Generalising sentences, on the other hand, express a pattern or regularity rather than a particular event or state [Krifka et al., 1995a]. In addition, there exist abstract entities which include Facts (objects of knowledge) and Propositions (objects of belief) and which can also have an SE type.

1.2 Annotation procedure

Annotation anchors

While [Friedrich et al., 2015] annotated situation entities on the level of automatically segmented clauses, we decided to consider the full verb of each clause as an anchor for annotation and to assign the situation entity types on top of these full verbs. Our decision is based on the observation that the automatic segmentation introduces many segmentation errors that have a negative impact on the quality of the data.

Infinitive clauses

Please note that infinitive clauses do not evoke a situation entity and are thus not annotated (see Example 6 where we only assign a label to vergaß but not to *füttern*).

(1) Pit füttert die Katzen.

(2) Pit hat die Katzen gefüttert.

(3) Pit wird die Katzen füttern.

(4) Pit möchte die Katzen füttern.

(5) Lasst uns die Katzen füttern.

(6) Pit vergaß, die Katzen zu füttern.

This means that for listings of infinitve clauses, as in the example below, we only label the finite main verb:

(7) Der Richter ist heute ein Mensch, der vieles gleichzeitig bewältigen muss: dem Zeugen **zuhören**, den Angeklagten **beobachten**, in der Akte **blättern**, sich die nächsten Fragen **überlegen**, nonverbale Singale aus der Verhandlung **mitnehmen** und zugleich all das noch vollständig und leserlich **notieren**,...

Annotation tool	As annotation interface, we use the INCEpTION[2] [Klie et al., 2018], a browser-based tool (see handout Introduction to INCEpTION).

[2] https://inception-project.github.io

2 Overview of the Classification Scheme

Table 1 gives an overview of our classification scheme, following previous work by [Friedrich et al., 2015, Mavridou, 2016].

Eventualities	State Event Event-Perfect-State - Report	**Kim** mag Katzen. **Kim** füttert die Katzen. **Kim** hat die Katzen schon gefüttert. ..., sagt Kim.
General Statives	Generalising Generic	**Kim** füttert oft meine Katze. **Katzen** sind immer hungrig.
Abstract Entities	Fact Proposition	**Ich** weiß, dass Kim die Katzen füttert. Ich glaube, dass Kim die Katzen füttert.
Speech acts	Question Imperative	Füttert **Kim** die Katzen? Vergiss nicht, die Katzen zu füttern!

Table 1: Overview of the classification scheme (bolded words are main referents, underlined words mark the anchor of the annotation).

3 Situation entity types

Here we give more details about the different types of situation entities (SE) and how they can be distinguished.

3.1 Main referent

In order to distinguish between different SE types, we first need to identify the *main referent* of a clause. The main referent of a clause is typically its *Topic* (i.e., what the clause is about).

[Krifka, 2007] describes *Topic* as follows:

> The topic constituent identifies the entity or set of entities under which the information expressed in the comment constituent should be stored in the Common Ground content.

We use the tests specified in [Krifka, 2007] to identify the main referent:

(8) An NP X is the Aboutness Topic of a sentence S containing X if
  a. S would be a natural continuation to the announcement
     Let me tell you something about X
  b. S would be a good answer to the question
     What about X?
  c. S could be naturally transformed into the sentence
     Concerning X, S'
     where S' differs from S only insofar as X has been replaced by a suitable pronoun.

Consider the following context:

> Sie können doch nicht im November Fahrzeuge anpreisen, wenn Sie in Ihren Gesetzentwurf reinschreiben, dass die künftig vielleicht auch nicht mehr fahren dürfen.

and the target sentence:

[Das]$_{Topic?}$ verstehen [die Menschen]$_{Topic?}$ draußen nicht.

If we apply the tests above, it becomes clear that "die Menschen", despite being the syntactic subject, are not the main referent of this clause.

There also exist sentences without a Topic, such as *tethic* sentences (Es regnet) and existentials Gott existiert/Es gibt gelbe Blumen/Es findet ein Markt statt).

Example 9a below behaves similarly to an existential and could be rephrased as Ex. 9b:

(9) a. Man hört in den letzten Tagen ja die merkwürdigsten Ausführungen.
b. Es gibt in den letzten Tagen ja die merkwürdigsten Ausführungen zu hören.

We annotate those sentence as STATE.

3.2 Genericity

An important concept for the annotation of SE types is genericity. To determine whether a clause is generic or not, we need to understand the following concepts: a) genericity of the *main referent* of the clause (i.e., *kind-referring NPs*) and b) genericity of the clause itself (*characterising sentences*). The examples below illustrate this difference.

(10) a. Der Dodo ist flugunfähig.
kind-referring, GENERIC
b. Der Dodo ist heute ausgestorben.
kind-referring, STATE
c. Der Dodo starb vor ca. 300 Jahren aus.
kind-referring, EVENT

Example 10a shows a generic sentence with a generic main referent. This is a *characterizing generic* that describes some property of instances of a kind. Example 10b, on the other hand, does not describe a property of instances of a kind but of the kind itself, and this property is temporally bound. A good test to tell the two apart is to insert a quantifying adverb:

(11) a. Der Dodo ist immer/meistens/oft flugunfähig.
b. *Der Dodo ist immer/meistens/oft ausgestorben.

We thus annotate Example 10a as GENERIC and Example 10b as a STATE. Example 10c also has a generic main referent, however, the clause refers to an event that took place at a certain point in time.

According to [Krifka et al., 1995b], kind-referring NPs "abstract away from particular objects (Examples 10a to 10c) , whereas with characterising sentences we abstract away from particular events and facts" (Example 10a) [Krifka et al., 1995b, p.4].

Table 2 gives more examples of (non-)generic main referents in (non-) generic clauses.

Table 3 shows that the form of the NP, i.e., whether it is definite or not, whether it contains quantification or the main referent is referenced by a proper name or not does not impact the genericity of the main referent.

To distinguish between the situation entitiy types GENERIC, GENERALISING, EVENT and STATE, we thus need to decide whether the main referent of the clause is kind-referring (or referring to a *well-established kind*; see below) or not and we need to determine whether the clause has a static, habitual or episodic reading.

clause / main referent	generic	non-generic
generic	Der Dodo ist flugunfähig. Der Löwe isst Fleisch.	Der Dodo starb vor 300 Jahren aus. Der Blobfisch wurde zum "hässlichsten Tier der Welt" gewählt.
non-generic	*nicht möglich*	Kim hätte gerne einen Dodo.

Table 2: Difference between generic (kind-referring) main referents and generic clauses/sentences.

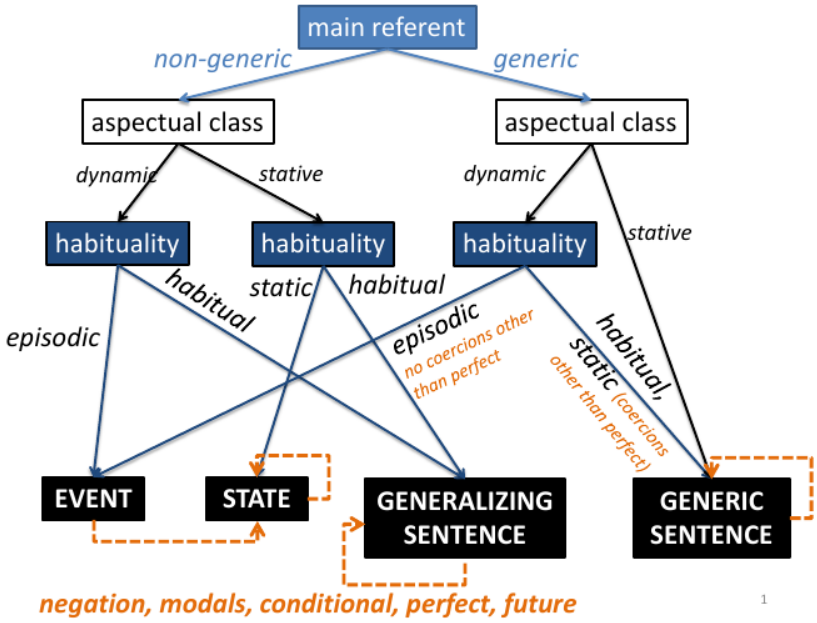

Figure 1: Decision tree for the following four situation entity types: GENERIC, GENERALISING, EVENT, STATE.

	kind-referring	**non-kind-referring**
definite NPs	Der Löwe ist ein Raubtier.	Der Löwe wurde in Brandenburg gesichtet.
indefinite NPs	Löwen essen Fleisch.	Löwen haben den Jeep angegriffen.
quantified NPs	Einige Wale sind bereits ausgestorben.	Einige Wale haben die Massenstrandung vor der Küste Tasmaniens überlebt.
proper names	Der Panthera leo persica, auch Persischer Löwe genannt, ist meist kleiner als der Afrikanische Löwe.	Captain Ahab mag keine Wale.

Table 3: Different forms of NPs and their genericity.

(12) a. **Der Löwe** ist ein Raubtier.
main referent: kind-referring
aspectual class: stative → GENERIC

b. **Simba** ist ein Raubtier.
main referent: non-kind-referring
aspectual class: stative
habituality: static → STATE

c. **Der Dodo** starb vor 300 Jahren aus.
main referent: kind-referring
aspectual class: dynamic
habituality: episodic → EVENT

d. **Kim** raucht gewöhnlich nach dem Essen.
main referent: non-kind-referring
aspectual class: stative
habituality: habitual → GENERALISING

3.2.1 Tests for Genericity

We now list some diagnostic tests for genericity, as detailed in [Krifka et al., 1995b, p.8-14]. Please note that the tests are only indicators but do not provide sufficient and necessary conditions for distinguishing between generic and non-generic situation entities.

Test 1: Characterising versus particular sentences

If we combine the clause with the adverbs *usually/typically* (in German: typischerweise/(für) gewöhnlich), then generic sentences should not (or only marginally) change their meaning (Examples 13a–13b) while for particular sentences the meaning will change considerably (Examples 14a–14b).

(13) a. Löwen essen Fleisch.
b. Löwen essen typischerweise/gewöhnlich Fleisch.

(14) a. Ein Löwe stand vor meinem Zelt.
b. Ein Löwe stand für gewöhnlich vor meinem Zelt.

Test 2: Kind-referring vs. non-kind-referring NPs

The second test in [Krifka et al., 1995b, p.10–11] tries to distinguish which types of NPs can be used as kind-referring. The basic idea here is

that NPs that can fill the subject position of predicates like *die out / be extinct* can be kind-referring while NPs that can fill the object position of predicates like *invent / exterminate* cannot be used as kind-referring (Examples 15a–15b). Example 16a shows that kind-referring NPs cannot fill the subject position of *invent*, as predicted.

This test, however, has a number of problems. Example 17a shows that not all non-kind-referring NPs can fill the object position of *invent* while there are NPs that can fill the subject slot of *die out* **and** the object slot of *invent* (Example 17b).

(15) a. Die Honigbiene wird bald aussterben. (kind-referring NP)
b. *Kim wird bald aussterben. (non-kind-referring NP)

(16) a. *Die Honigbiene wurde im 17. Jahrhundert erfunden.
b. Fortnite wurde 2011 von einer Firma namens Epic Games erfunden.

(17) a. *Kim wurde von ihren Eltern erfunden.
b. Nein das Spiel Fortnite wird nicht aussterben.

Test 2 Characterising vs. particular sentences[3]

Sentences that exclude stative readings, like Example 18b, do not have a characterising reading. While Standard German grammar does not have a progressive tense, in colloquial German the "Rheinische Verlaufsform" provides the same function (Example 19b)

(18) a. The Italian drinks wine with his dinner. characterising
b. The Italian **is drinking** wine with his dinner. particular

(19) a. Der Italiener trinkt Wein zum Essen. characterising
b. Der Italiener war während des Essens **am Wein trinken**. particular

Example 18b also illustrates that NPs that can have a kind-referring reading can also occur in non-generic sentences.

Test 3: Kind (and well-established kind)-referring vs. non-kind-referring NPs

The next test focusses on the distinction between **kind-referring** (or *well-established kind*-referring) clauses and those that refer to specific individuals and objects in the world. What constitutes "a well-established kind" is by no means clear and [Krifka et al., 1995b, p.11] admit that they do not have a clear answer to this question.[4]

The authors illustrate the distinction with the examples below, stating that the Coke bottle constitutes a well-established kind and thus can have a generic reading while the green bottle does not, and therefore the main referent must have an object-referring reading:

(20) a. The Coke bottle has a narrow neck.
b. ???The green bottle has a narrow neck.

[3] This is test 4 in [Krifka et al., 1995b, p.12].
[4] See footnote 9 in [Krifka et al., 1995b, p.11] where the authors state that they "will not attempt to offer any sort of analysis of the notion 'well-established kind'."

A well-established kind Below we try to provide some examples and guidelines for identifying instances of *a well-established kind*.

According to the Historical Dictionary of Biology, a *kind* is defined as

> [...] a set of organisms that are similar in one or more respects or that form a cohesive unit. The grouping may be based on the fact that the organisms are similar in their shape and behavior [...][5]

Based on this definition, we consider a *well-established kind* to be

> *a cohesive unit of entities that are similar to each other in one or more respects.*

This perception of entities as "similar" does not need to be based on similarity in the real world but can also be a perceived similarity or an attributed affiliation to a specific group, based on stereotypes and biases. Here it is important that the entities are not only considered as a cohesive group by some individuals but that this view is shared amongst a larger group of people.

Stereotypes Also relevant for our definition of a *well-established kind* is the concept of *stereotypes*. According to Wikipedia, a stereotype is defined as:

> [...] eine im Alltagswissen präsente Beschreibung von Personen oder Gruppen, die einprägsam und bildhaft ist: Sie vereinfacht einen als typisch behaupteten Sachverhalt und bezieht sich auf diesen. Stereotype sind gleichzeitig relativ starre, überindividuell geltende beziehungsweise weit verbreitete Vorstellungsbilder.[6]

A reference to *a well-established kind* denotes a group of entities that is associated with certain expectations regarding their appearance, behaviour etc. Therefore, only Example 21a can be considered as kind-referring while Example 21b does not trigger any expectation bias regarding the members of the group ("people that have a last name starting with an "R" should cue over here."). Example 21c is ambiguous and could potentially refer to the zodiac sign "Scorpion", and thus raise the expectation that members of this group display certain characteristics.

(21) a. Menschen, die einen Opel Manta fahren ✓
b. Menschen, deren Nachname mit "R" anfängt ✗
c. Menschen, die im November Geburtstag haben ???

To test whether a particular NP X qualifies as a reference to a well-established kind, we propose to use the same tests as outlinend above.

Test If X can fill the subject slot of predicates like "aussterben", then it can be considered as *referring to a well-established kind*.

(22) Some examples:
a. Tierärzte: "Die deutsche Nutztierhaltung wird bald aussterben".
b. Der Homo Bürohocker wird bald aussterben.
c. Der Ruhrpott-Proll wird nie aussterben.

[5] See `https://begriffsgeschichte.de/doku.php/begriffe/art`.
[6] See `https://de.wikipedia.org/wiki/Stereotyp`.

Main referent	**Examples**	kind-referring
Organisations	IBM, SPD, Bundestag	✗
Appearance	Liliputaner, Weiße	✓
Behaviour	Sonntagsfahrer, Hooligans, Mantafahrer	✓
Origin	die Deutschtürken, der Ruhrpott-Proll	✓
Culture	die Kurden	✓
Religion	die Christen, die Moslems, Kirchengänger	✓
Tradition	Der typische Oktoberfestgänger	✓
Language	Texasdeutsch-Sprechende	✓
Nationality	die Deutschen, die Franzosen	✓
Social Class	Prolls, Reiche, Mittelklasse	✓
Age	Teenager, alte weiße Männer	✓
Gender	Frauen, Männer, Transfrau	✓
Sexual identity	Lesben, Schwule	✓
Institution	Studierende, Bänker	✓
Berufe	die Lehrerinnen, Kindergärtner, Pfarrer	✓
Brands	Fortnite, Tempo-Taschentuch, Colaflasche	✓
Politik	Kommunisten, Nazis, Terroristen	✓
Sport	Handballerinnen, Boxer, Marathonläufer	✓
Placeholder	Erika Mustermann, Otto Normalverbraucher, Lieschen Müller	✓

Table 4: Attributes that are indicative of *a (well-established) kind*.

d. Wird Fortnite aussterben?
(referring to the gaming culture of Fortnite players)

> Please note that this test does not specify whether or not the NP has a generic reading in context, but only that it *might* have a generic reading, given the right context.

Table 4 lists examples for different types of groups / attributes for main referents, and whether we consider them as *potentially* kind-referring or not.

Organisations

While we do not consider references to specific organisations (Siemens, BMW, Greenpeace, North Korea etc.) as kind-referring, we do allow for kind-referring readings for groups of organisations. Consider Examples 23a-23b. Here, specific entities have been selected and are assigned to a group with certain ascribed properties and expectations regarding their behaviour.

(23) a. Internetunternehmen wie Google, Twitter, Tinder, Facebook ✓

b. autokratische Staaten mit Atomwaffen wie Nordkorea, der Iran, oder Pakistan ✓

Brands

Brand names can also function as kind-referring, as shown in Examples (24a)-(24b), but can also have a non-generic reading, as in (24c).

(24) a. Das Tempo-Taschentuch hat 3 Lagen und ist äußerst reißfest.✓

b. Seit über 40 Jahren steht der 3er BMW wie kein anderes Fahrzeug für dynamisches Fahrvergnügen.✓

c. Der BMW steht nicht mehr vorm Haus. ✗

Placeholder We also treat entities like "der kleine Handwerker" and "Oma Erna" as kind-referring (see Example 25a), similar to other placeholder names. This, however, does not apply to the use of examples like 25b. Those instances are not considered as kind-referring but as a reference to a specific person that might or might not exist in real life.

(25) a. Aber wehe, der kleine Handwerker tauscht sich mit seinem Kollegen auf der Baustelle über WhatsApp aus oder Oma Erna stellt die Fotos von der letzten Kaffeefahrt auf Facebook ✓

b. Dann ist da diese ältere komische Tante, die wieder nur sich selbst zum Fest mitgebracht hat ... ✗

Test If we can insert "typischerweise/(für) gewöhnlich", then the sentence has a characterising reading.

(26) Some examples:

a. Sonntagsfahrer stehen auf dem Londoner Boden in üblem Ruf.

b. Sonntagsfahrer stehen **für gewöhnlich** auf dem Londoner Boden in üblem Ruf. → GENERIC

(27) a. Der Lehrer hat vormittags recht und nachmittags frei.

b. Der Lehrer hat **für gewöhnlich** vormittags recht und nachmittags frei. → GENERIC

(28) a. Ihr Mann war Lehrer wie sie selbst.

b. ???Ihr Mann war **für gewöhnlich** Lehrer wie sie selbst. → *non-generic,* STATE

(29) a. An deutschen Schulen fehlen mindestens 20,000 Lehrer.

b. ?An deutschen Schulen fehlen **für gewöhnlich** mindestens 20,000 Lehrer. → *different reading; non-generic,* STATE

(30) a. Wie keine ihrer Nachfolgerinnen erlangte diese Erika Mustermann ungewöhnlichen Ruhm.

b. Wie keine ihrer Nachfolgerinnen erlangte diese Erika Mustermann **für gewöhnlich** ungewöhnlichen Ruhm. → *different reading; non-generic, ambiguous*

Indefinite pronouns

Below we provide examples how to deal with indefinite pronouns like *wer, jeder, alle, man, ...* *Wer, jeder, alle* here refer to individual entities characterised as "having visions". We thus consider those as non-generic. *Man*, on the other hand, does not refer to specific entities and is treated as a non-generic main referent.

1. **Wer** Visionen hat, sollte zum Arzt gehen.→ *non-generic*, STATE
2. Jeder, **der** Visionen hat, sollte zum Arzt gehen.→ *non-generic*, STATE

3. Wenn **man** Visionen hat, sollte man zum Arzt gehen.→ *generic*, GENERIC
4. Alle, **die** Visionen haben, sollten zum Arzt gehen. → *non-generic*, STATE
5. Wenn **Kim** Visionen hat, sollte sie zum Arzt gehen.→ *non-generic*, STATE

This, however, does not necessarily apply to the whole subject NP. As before, we need to determine whether the main referent of the sentence is kind-refering or not. In the examples below, the subject NPs describe instances of a well-established kind.

For the predicate of the main sentence, we also annotate the generic reading for Examples 1-4. For the non-generic main referent in Example 5, we get again the stative reading.

1. **Wer Visionen hat**, sollte zum Arzt gehen.→ GENERIC
2. **Jeder, der Visionen hat**, sollte zum Arzt gehen.→ GENERIC
3. Wenn man Visionen hat, sollte **man** zum Arzt gehen.→ GENERIC
4. **Alle, die Visionen haben**, sollten zum Arzt gehen. → GENERIC

The next three examples show variations of the same sentence with a non-generic main referent. For the relative clause (Example 1), we annotated STATE while the main clause describes an EVENT (Example 2). When the sentence is expressed as a conditional, as in Example 3, then the EVENT is coerced into a STATE.

1. Kim, **die** Visionen hat, ging gestern zum Arzt. → STATE
2. **Kim**, die Visionen hat, ging gestern zum Arzt. → EVENT
3. Wenn Kim Visionen hat, sollte **sie** zum Arzt gehen.→ STATE
4. **Wer da von einem wichtigen Beitrag Deutschlands zur Senkung der CO^2-Emissionen weltweit redet**, will oder kann nicht rechnen. → STATE

Example 4, on the other hand, refers to a group of people (everybody who is claiming that Germany makes an important contribution for decreasing the CO^2 emissions worldwide). This group, however, is neither a kind nor a well-established kind but a group of specific individuals. We therefore annotate this sentence as STATE.

Conditionals

Conditionals are hypothetical conditions that describe possible events.

Here it is not important whether the event has already taken place (a), or might happen some time in the future (b). In (c) it is unclear whether a purely hypothetical condition is being described or whether the utterance describes an observed behavior that has been taken place in the real world. In the last example, (d), it is clear that the described event is not hypothetical but has already taken place.

In all the examples below, we annotate STATE.

(a) Wenn also **die nukleare Abschreckung** den Frieden erhalten hat, → *non-generic* STATE
darf auch in unserer Zeit **die Glaubwürdigkeit der nuklearen Abschreckung** nicht gefährdet werden. → *non-generic* STATE

(b) Wenn **wir** dieses Gesetz auf den Weg bringen, → *non-generic* STATE
wird **es** ein Meilenstein auf dem Weg zu mehr Klimagerechtigkeit sein.
→ *non-generic* STATE

(c) Wenn **man** sich nur auf die Braunkohle konzentriert, → *generic*
GENERIC zeichnet **man** ein falsches Bild. → *generic* GENERIC

(d) Da kann **man** sich, ehrlich gesagt, doch nur an den Kopf fassen,
→ *generic* STATE
wenn sich in Bonn **über 20 Staaten** zusammentun,
→ *non-generic* STATE
eine Allianz für den Kohleausstieg gründen → *non-generic* STATE
und **Deutschland** wie ein begossener Pudel an der Seite steht.
→ *non-generic* STATE

3.3 Generalising

Repeated events that describe a habitual or characterising feature of a non-generic main referent are annotated as GENERALISING:

(31) Dafür habe ich mich auch als Klimapolitikerin immer wieder stark gemacht.

In the example above, it is a characterising feature of the speaker that she is a strong advocate of X (dafür).

3.4 Report

REPORT, a subtype of EVENT, is introduced by verbs of speech. To count as a verb of speech and to be annotated as REPORT, a clause needs to include a message. Compare the following examples below:

(32) Kim sagte, dass morgen die Sonne scheint. REPORT

(33) Kim sagte : morgen scheint die Sonne. REPORT

(34) Kim bezeichnet den Klimawandel als eine zentrale Herausforderung. REPORT

(35) Kim sprach übers Wetter. EVENT

(36) Kim erzählte wieder Geschichten. EVENT

(37) Kim sagte mir, warum sie keine Katzen mag. EVENT

Examples 32–34 include a message (something that is being reported) while Examples 35–37 merely describe speech events and their topic, but not the message. For example, in Ex. 37 the listener only learns that Kim told somebody why she doesn't like cats but doesn't get to hear the message, i.e., why Kim doesn't like cats.

In the context of future tense, conditionals and modals expressing possiblility or necessity, we also assume coercion for the REPORT label (see examples below):

(38) Die Bundesregierung **sagt**, dass eine landesweite Impfkampagne geplant ist. REPORT

(39) Die Bundesregierung wird später **sagen**, dass das alles geplant war. STATE

(40) Wenn die Bundesregierung **sagt**, dass das alles geplant war, ... STATE

(41) Sie könnten den Bürgern **sagen**, dass eine landesweite Impfkampagne geplant ist. STATE

(42) Sie müssen den Bürgern **sagen**, dass eine landesweite Impfkampagne geplant ist. STATE

The following examples 44-45 show some ambiguous cases that might be considered as either REPORTor EVENT. While the direct speech in Ex.43 can be easily identified as REPORT, the placeholder "so" in Ex. 44 is similar to Ex. 45 and thus questions this analysis. We follow the annotation that we would choose for Ex. 45 and annotate both instances as EVENT.

(43) "One for all and all for one", sagte **Stoltenberg**. REPORT

(44) "One for all and all for one", so sagte es **der NATO-Generalsekretär Stoltenberg** anlässlich der Feierlichkeiten. EVENT

(45) "One for all and all for one". Das sagte **Stoltenberg**. EVENT

3.5 Abstract Entities

Direct/indirect speech

We annotate clauses as FACT if they are presented as something known to be true, and as PROPOSITION if the speaker is not fully commited to the truth value of the clause (e.g., if it is believed to be true or if the speaker is unsure or expresses speculation):

(46) Sie weiß, dass wir das schaffen. FACT

(47) Sie weiß: Wir schaffen das. FACT

(48) Ich glaube, wir schaffen das. PROPOSITION

(49) Ich glaube: Wir schaffen das. PROPOSITION

(50) Ich hoffe, dass dieses Instrument jetzt wirklich kommt. PROPOSITION

(51) Ich denke, dass es dem Klima egal ist, wo auf der Welt welche Tonne CO^2 in die Luft geblasen wird. PROPOSITION

The crucial point is not whether the embedded clause is true or false but whether the speaker is commited to the truth value of the proposition.

Test

Factive verbs cannot be introduced with "I'm not sure but...":

(52) Ich weiß es nicht sicher, aber ich habe gehört, dass sie da war.

(53) *Ich weiß es nicht sicher, aber ich habe gesehen, dass sie da war.

(54) *Ich weiß es nicht sicher, aber ich habe erfahren, dass sie da war.

Here are some more examples for FACT:

FACT

Kim weiß,	dass Pit verreist ist.
Kim hat gesehen,	dass Pit abgereist ist.
Kim realisierte,	dass Pit verreist war.
Kim hat erfahren,	dass Pit verreist ist.
Kim hat gehört,	dass Pit verreist ist.

Fact		Proposition	
bedauern	dass ...	ahnen	dass ...
bemerken		andeuten	
beobachten		annehmen	
bereuen		anzweifeln	
entdecken		denken	
entschuldigen		fürchten	
erfahren		glauben	
erinnern		hoffen	
merken		implizieren	
mitbekommen		inferieren	
realisieren		meinen	
sehen		mutmaßen	
übelnehmen		prophezeien	
vergessen		schlussfolgern	
wahrnehmen		vermuten	
wissen		vorstellen	
zeigen		zweifeln	
bestürzt sein	dass ...	einig sein	dass ...
dankbar sein		einleuchtend sein	
empört sein		gewiss sein	
erfreut sein		klar sein	
schockiert sein		offensichtlich sein	
überrascht sein		sicher sein	
verwundert sein		sonnenklar sein	
zufrieden sein		überzeugt sein	
Freude empfinden	dass ...	Angst haben	dass ...
Reue zeigen		der Hoffnung sein	
...		...	

Table 5: Non-exhaustive list of expressions that we annotate as Fact (left) and Proposition (right).

And here are some more examples for Proposition:

Proposition	
Kim glaubt,	dass Pit verreist ist.
Kim vermutet,	dass Pit verreist ist.
Kim denkt,	dass Pit verreist ist.
Kim nimmt an,	dass Pit verreist ist.
Kim ist überzeugt,	dass Pit verreist ist.
Kim zweifelt daran,	dass Pit verreist ist.

Some verbs additionally encode the speaker's attitude towards the proposition:

(55) Kim hofft, dass Pit verreist ist.

(56) Kim wünscht sich, dass Pit verreist ist.

(57) Kim fürchtet, dass Pit verreist ist.

Our annotation scheme does not encode the speaker's attitude / the desirability of the clause.

3.5.1 Fact/Proposition in conditionals and questions

(58) Henry discovered that Louise is in NY.
presuppositional and factive, Fact

In the context of a conditional construction, question or with an epistemic adverb (see examples below), the above clause can loose its presuppositional meaning and it is now possible that Louise either is or is not in NY.

Coercion (Fact ⇒ non-factive: Proposition)

(59) If Henry discovers that Louise is in NY, he'll be furious. Proposition

(60) Why is Henry in such a bad mood? Did he discover that Louise is in NY? Proposition

(61) Henry is in a terrible mood. Perhaps he's discovered that Louise is in NY. Proposition

3.6 Imperatives

Verbs in the imperative mood are annotated as Imperative.

(62) Machen Sie der Union Druck! Imperative

(63) Passen Sie auf, dass Sie nicht zum Sargnagel Europas werden.
Imperative

The example below shows an imperative clause with a generic main referent (referring to the kind of people who want to enter into a lifetime relationship). We annotate those clauses as Imperative, not as Generic.

(64) Darum <u>prüfe</u>, wer sich ewig bindet... Imperative

4 Coercion

Some linguistic phenomena can change the situation entity type of events. Examples are negation, future tense, modals, conditionals, and subjectivity (also see §4.3 in the Situation Entity Guidelines of [Friedrich et al., 2015]). Below we give examples for each phenomenon.

4.1 Future tense

The following two examples describe potential events that might take place in the future. This causes coercion (in Example 1 from Event to Generic, in Example 2 from Event to State). The label difference is due to *wer* in Ex. 1 referring to an unspecific group of humans (thus *generic*) while Ex. 2 is referring to specific individuals that might become chancellor (thus *non-generic*).

1. Egal **wer** <u>regieren</u> wird, er darf nicht versagen. → *generic* Generic
2. Es ist mir egal, **wer** unter mir Bundeskanzler <u>wird</u>. → *non-generic* State

4.2 Modals

The next examples illustrate the coercion from EVENT to either STATE or GENERIC, depending on the features of the main referent (either non-generic or *generic*):

1. **Kim** protestiert heute gegen den Klimawandel. → *non-generic* EVENT
2. **Kim** möchte gegen den Klimawandel protestieren. → *non-generic* STATE(coercion)
3. **Wer** in unserer Gesellschaft gegen den Klimawandel protestieren will, ... → *non-generic* STATE (coercion)
4. Jeder, **der** in unserer Gesellschaft gegen den Klimawandel protestieren will, ... → *non-generic* STATE (coercion)
5. Alle, **die** in unserer Gesellschaft gegen den Klimawandel protestieren wollen, ... → *non-generic* STATE (coercion)

Please note that for examples like the one below where *dürfen* doesn't have a deontic reading, we do not assume coercion but annotate the situation type as EVENT:

(65) Frau Staatssekretärin, ich darf Sie bitten! EVENT

5 Specific phenomena

5.1 Expletives

For expletives, identifying the main referent is not always easy. In German, different uses of expletives have to be distinguished:

1. Formal subject or object without semantic content
   Es regnet $\emptyset_{main\ referent}$.
2. Correlate of an extraposed clausal argument
   Hier bringt es wenig, $\underline{\textit{Bewerbungen herumzuschicken}}_{main\ referent}$
3. Vorfeld-es (prefield *es*)
   Es wird $\underline{\text{viel}}_{main\ referent}$ schöngeredet, und es passiert $\underline{\text{nichts}}_{main\ referent}$.

For the first case (expletive *es* as formal subject/object), no main referent exists. For the placeholder uses (correlate and prefield *es*), the main referents are underlined.

5.2 Quantifiers

Quantifiers such as *jeder/keiner/alle/niemand/...* do not have an effect on the genericity of the main referent (see examples below):

(66) a. Alle Menschen sind sterblich. *kind-referring*, GENERIC

b. Kein Mensch ist unsterblich. *kind-referring*, GENERIC

c. Kein Schwein ruft mich an, keine Sau interessiert sich für mich. *non-kind-referring*, STATE

d. Normalerweise ruft Sonntags keine Sau an. *non-kind-referring*, GENERALISING

e. Alle Staaten der Welt sind an Bord.
non-kind-referring, STATE

Negation is a bit tricky (see Example 66b). We assume that the intended reading for this sentence is:

(67) "For any X: if X belongs to mankind, then X is mortal."

or:

(68) "There is no X for which holds: X belongs to mankind AND X is mortal."

Both statements, (66a) and (66b), refer to instances of the same kind. We thus consider negated references to kinds as *kind-referring*. In Example 66c, however, the referent is non-generic and (most probably) only refers to the group of individuals known to the speaker. Thus, we consider this negated example as non-kind-referring and annotate the clause as STATE.

Example 66d illustrates that in some contexts the clause can get a habitual reading.

When used as a substituitive personal pronoun, the NP/clause is again ambiguous and has to be resolved in context. Please note that negation can cause coercion of EVENT to STATE (see Examples 69g-69h).

(69)
a. Jeder ist sterblich. *kind-referring*, GENERIC
b. Jeder hat schon ausgetrunken, nur Kims Glas ist noch halbvoll. *non-kind-referring*, STATE
c. Alle sind sterblich. *kind-referring*, GENERIC
d. Alle sind an Bord. *non-kind-referring*, STATE
e. Niemand ist unsterblich. *kind-referring*, GENERIC
f. Niemand will eine Mauer errichten. *non-kind-referring*, STATE
g. Kim hat behauptet, dass das auch vor dem Hintergrund der Erreichung unserer Klimaziele leicht werden wird. *non-kind-referring*, EVENT
h. Keiner hat behauptet, dass das auch vor dem Hintergrund der Erreichung unserer Klimaziele leicht werden wird. *non-kind-referring*, STATE

5.3 Foreign language material

Our data also includes some foreign language material (mostly English). We do not mark those utterances as FOREIGN LANGUAGE MATERIAL but annotate them, same as the German data.

5.4 Formulaic expressions

We do not annotate formulaic expressions like exclamations used to express surprise, to congratulate, thank or apologise. We also do no

Examples for exclamations that do not describe situations:

(70) Oh mein Gott!

(71) Herzlichen Glückwunsch!

(72) Vielen Dank!

(73) Entschuldigung!

We also consider the following examples of idiosyncratic expressions as non-situation entities as they can often be replaced by an adjective or an adverb of manner.

(74) Ich möchte den Bericht **wie folgt** wiedergeben.

(75) Ich möchte den Bericht **folgendermaßen** wiedergeben.

(76) Ich möchte den Bericht **stark gekürzt** wiedergeben.

(77) Die jungen Leute haben gefeiert, **was das Zeug hält**.

(78) Die jungen Leute haben **ausgiebig** gefeiert .

(79) Die jungen Leute haben gefeiert **wie verrückt**.

(80) Die jungen Leute haben gefeiert **bis zum Abwinken**.

(81) Die jungen Leute haben gefeiert, **bis der Arzt kommt**.

5.5 Agreement / Rejection / Commitments /...

Sometimes it is not quite clear whether a clause that makes a proclamation or expresses rejection should be considered as an EVENT or as a STATE. Here it depends whether the utterance in context is focussed on the event (as in Ex. 84) or on the speaker's attitude (as in Ex. 83).

(82) Folglich lehnen wir diesen Antrag ab. STATE

(83) Folglich lehnen wir diesen Antrag ab, da er mit unseren Werten unvereinbar ist. STATE

(84) Folglich haben wir diesen Antrag in der Abstimmung abgelehnt. EVENT

Here are some more examples:

(85) Deshalb bekennt sich die Bundesrepublik Deutschland ausdrücklich zu einer Welt ohne Atomwaffen. STATE

5.6 Participle or adjective

The distinction between adjective and participle is not always clear. Here we provide a list of ambiguous instances and document whether we consider them as a participle (and annotate them) or as an adjective (no annotation):

word form	**verb**	**adjective**	**example**
verpflichtet	✓		Wir haben uns verpflichtet, ...

5.7 Label conflict due to coordination

For modal constructions, we do not label the modal verb but assign the SE label to the infinitive verb. Due to VP coordination, we can encounter situations where we have more than one modal with different main referents (Example below: *man* generic, *wir* non-generic) but only one infinitive verb form (*differenzieren*). Here we assign the label that best reflects the reading of the whole clause in context.

(86) Allerdings kann man und wollen wir differenzieren

6 Specific verbs

Here we provide guidelines for the annotation of specific verbs that are frequent in our data and where the annotation is not straightforward.

6.1 lassen

For the verb *lassen*, it is not always clear whether to consider it as an auxiliary and anchor the annotation on the verbal dependent or whether to consider *lassen* itself as the main verb. Here we need to distinguish between different readings:

(87) Imperativformen (Annotation auf *lassen*)
  a. Lass das!
  b. Du sollst das lassen.

(88) Zustandsformen mit Partizip (Annotation auf *lassen*)
  a. Ich ließ das Auto in der Garage (stehn) und ging zu Fuß.
  b. Ich ließ den Fernseher aus und nahm ein Buch.
  c. Sie lassen ihre Tochter nach 10 noch auf der Straße spielen.

(89) Reflexivkonstruktion mit Infinitiv (Annotation auf dem *Vollverb*)
  a. Das lässt sich leicht realisieren.
  b. Das Fahrrad lässt sich noch reparieren.
  c. Ich lass mir meine Zähne richten.
  d. Lasst uns das tun.

Weitere Beispiele (noch nicht klassifiziert):

(90) a. Ich lass mir meine Zähne richten.
  b. Lasst uns das tun.

6.2 zeigen/beweisen

The verbs *show/prove* are frequent in our data and are ambiguous between a stative and an eventive reading. We follow [Mavridou, 2016] and annotate those as State:

(91) Die bekannte Wahrheit zeigt sich auch hier *non-generic*, State

(92) Die aggressive Außenpolitik von Russland und Nordkorea beweist, dass ... *non-generic*, State

6.3 zulassen

Contrary to [Mavridou, 2016], we do not consider *zulassen* as an EVENT but annotate it as STATE.

(93) Sie ließ nicht zu, dass X STATE

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